

Aura AI - System Design: Device Migration Strategy

What is Backed Up

User profile, chat history, personality traits, reminders, application settings, and AI persona configuration are backed up. Temporary cache files, logs, and session tokens are excluded.

Backup Location

Encrypted backups are stored in secure cloud storage linked to the user's account, such as Google Drive or a dedicated backend service.

File Format and Encryption

Data is exported into a structured JSON backup and encrypted using AES-256. Encryption keys are managed through Android Keystore to protect data at rest and in transit.

Restore Flow

1. User installs the app on a new device. 2. User signs into the same account. 3. Backup is detected and downloaded. 4. Data is decrypted on-device. 5. Room Database and DataStore are restored. 6. SyncManager validates and synchronizes missing changes. 7. User continues with complete history, settings, and persona.

Conflict Resolution

Local wins strategy: the most recent local modifications are preserved and synchronized during the next sync cycle.

Intentionally Excluded

Cache files, temporary downloads, audio recordings, logs, and session tokens are excluded because they can be regenerated or may contain sensitive information.